

```
In [6]: pip install ipython-sql sqlalchemy pyodbc
```

```
Requirement already satisfied: ipython-sql in c:\users\mgts\anaconda3\lib\site-packa
ges (0.5.0)
Requirement already satisfied: sqlalchemy in c:\users\mgts\anaconda3\lib\site-packag
es (2.0.39)
Requirement already satisfied: pyodbc in c:\users\mgts\anaconda3\lib\site-packages
(5.2.0)
Requirement already satisfied: prettytable in c:\users\mgts\anaconda3\lib\site-packa
ges (from ipython-sql) (3.9.0)
Requirement already satisfied: ipython in c:\users\mgts\anaconda3\lib\site-packages
(from ipython-sql) (8.30.0)
Requirement already satisfied: sqlparse in c:\users\mgts\anaconda3\lib\site-packages
(from ipython-sql) (0.5.5)
Requirement already satisfied: six in c:\users\mgts\anaconda3\lib\site-packages (fro
m ipython-sql) (1.17.0)
Requirement already satisfied: ipython-genutils in c:\users\mgts\anaconda3\lib\site-
packages (from ipython-sql) (0.2.0)
Requirement already satisfied: greenlet!=0.4.17 in c:\users\mgts\anaconda3\lib\site-
packages (from sqlalchemy) (3.1.1)
Requirement already satisfied: typing-extensions>=4.6.0 in c:\users\mgts\anaconda3\l
ib\site-packages (from sqlalchemy) (4.12.2)
Requirement already satisfied: decorator in c:\users\mgts\anaconda3\lib\site-package
s (from ipython->ipython-sql) (5.1.1)
Requirement already satisfied: jedi>=0.16 in c:\users\mgts\anaconda3\lib\site-packag
es (from ipython->ipython-sql) (0.19.2)
Requirement already satisfied: matplotlib-inline in c:\users\mgts\anaconda3\lib\site
-packages (from ipython->ipython-sql) (0.1.6)
Requirement already satisfied: prompt-toolkit<3.1.0,>=3.0.41 in c:\users\mgts\anacon
da3\lib\site-packages (from ipython->ipython-sql) (3.0.43)
Requirement already satisfied: pygments>=2.4.0 in c:\users\mgts\anaconda3\lib\site-p
ackages (from ipython->ipython-sql) (2.19.1)
Requirement already satisfied: stack-data in c:\users\mgts\anaconda3\lib\site-packag
es (from ipython->ipython-sql) (0.2.0)
Requirement already satisfied: traitlets>=5.13.0 in c:\users\mgts\anaconda3\lib\site
-packages (from ipython->ipython-sql) (5.14.3)
Requirement already satisfied: colorama in c:\users\mgts\anaconda3\lib\site-packages
(from ipython->ipython-sql) (0.4.6)
Requirement already satisfied: wcwidth in c:\users\mgts\anaconda3\lib\site-packages
(from prompt-toolkit<3.1.0,>=3.0.41->ipython->ipython-sql) (0.2.5)
Requirement already satisfied: parso<0.9.0,>=0.8.4 in c:\users\mgts\anaconda3\lib\si
te-packages (from jedi>=0.16->ipython->ipython-sql) (0.8.4)
Requirement already satisfied: executing in c:\users\mgts\anaconda3\lib\site-package
s (from stack-data->ipython->ipython-sql) (0.8.3)
Requirement already satisfied: asttokens in c:\users\mgts\anaconda3\lib\site-package
s (from stack-data->ipython->ipython-sql) (3.0.0)
Requirement already satisfied: pure-eval in c:\users\mgts\anaconda3\lib\site-package
s (from stack-data->ipython->ipython-sql) (0.2.2)
Note: you may need to restart the kernel to use updated packages.
```

```
In [7]: %reload_ext sql
```

```
In [14]: %sql mssql+pyodbc://@AAAA\MSSQLSERVER02/Amira?driver=ODBC+Driver+17+for+SQL+Server&
```

```
In [9]: %config SqlMagic.style = 'DEFAULT'
```

EDA

1- Sales by Country

2- Country Segmentation

```
In [10]: %%sql
select top(5) * from dbo.adding

* mssql+pyodbc://@AAAA\MSSQLSERVER02/Amira?driver=ODBC+Driver+17+for+SQL+Server&tr
sted_connection=yes
Done.
```

```
Out[10]: InvoiceNo  StockCode  Description  Quantity  InvoiceDate  UnitPrice  CustomerID  Country
```

536365	85123A	WHITE HANGING HEART T- LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	17850	Unitec Kingdom
536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	17850	Unitec Kingdom
536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	17850	Unitec Kingdom
536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39	17850	Unitec Kingdom
536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-01 08:26:00	3.39	17850	Unitec Kingdom

Creating a Country Segmentation View

```
In [11]: %%sql

create view Country_Level as
select country, round(sum( Quantity* UnitPrice), 0) total_sales,
```

```

case
    when sum( Quantity* UnitPrice)> 100000 then 'vip country'
    when sum( Quantity* UnitPrice)> 50000 then 'average country'
    else 'Normal country'
end as Country_Levels

from dbo.adding
group by country
order by total_sales desc

```

```

* mssql+pyodbc://@AAAA\MSSQLSERVER02/Amira?driver=ODBC+Driver+17+for+SQL+Server&trusted_connection=yes
(pyodbc.ProgrammingError) ('42000', '[42000] [Microsoft][ODBC Driver 17 for SQL Server][SQL Server]The ORDER BY clause is invalid in views, inline functions, derived tables, subqueries, and common table expressions, unless TOP, OFFSET or FOR XML is also specified. (1033) (SQLExecDirectW)')
[SQL: create view Country_Level as
select country, round(sum( Quantity* UnitPrice), 0) total_sales,
case
    when sum( Quantity* UnitPrice)> 100000 then 'vip country'
    when sum( Quantity* UnitPrice)> 50000 then 'average country'
    else 'Normal country'
end as Country_Levels

from dbo.adding
group by country
order by total_sales desc]
(Background on this error at: https://sqlalche.me/e/20/f405)

```

Sales by Country

In [12]: %%sql

```
select top(5)* from Country_level
```

```

* mssql+pyodbc://@AAAA\MSSQLSERVER02/Amira?driver=ODBC+Driver+17+for+SQL+Server&trusted_connection=yes
Done.

```

Out[12]:

country	total_sales	Country_Levels
Australia	137077.0	vip country
Austria	10154.0	Normal country
Bahrain	548.0	Normal country
Belgium	40911.0	Normal country
Brazil	1144.0	Normal country

Country Segmentation

In [13]: %%sql

```
select Country_Levels, count(country) as country_count
```

```
from Country_Level  
group by Country_Levels  
order by country_count desc
```

```
* mssql+pyodbc://@AAAA\MSSQLSERVER02/Amira?driver=ODBC+Driver+17+for+SQL+Server&tru  
sted_connection=yes  
Done.
```

Out[13]:

Country_Levels	country_count
Normal country	30
vip country	6
average country	2

In []: